

homework-2

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Question 1

```
install.packages("aRtsy")
```

The function “install.packages()” lets you download pre-made functions and datasets for use in future sessions. Once you install them, you won’t need to do it again.

The function “library()” loads the installed package in your current R session. It’s similar to checking the box next to the package you want to use in the Packages tab. You might want to dedicate a code chunk to all the packages you’ll use so your document renders properly.

Packages

Question 2

```
library(aRtsy)
```

Collatz Conjecture:

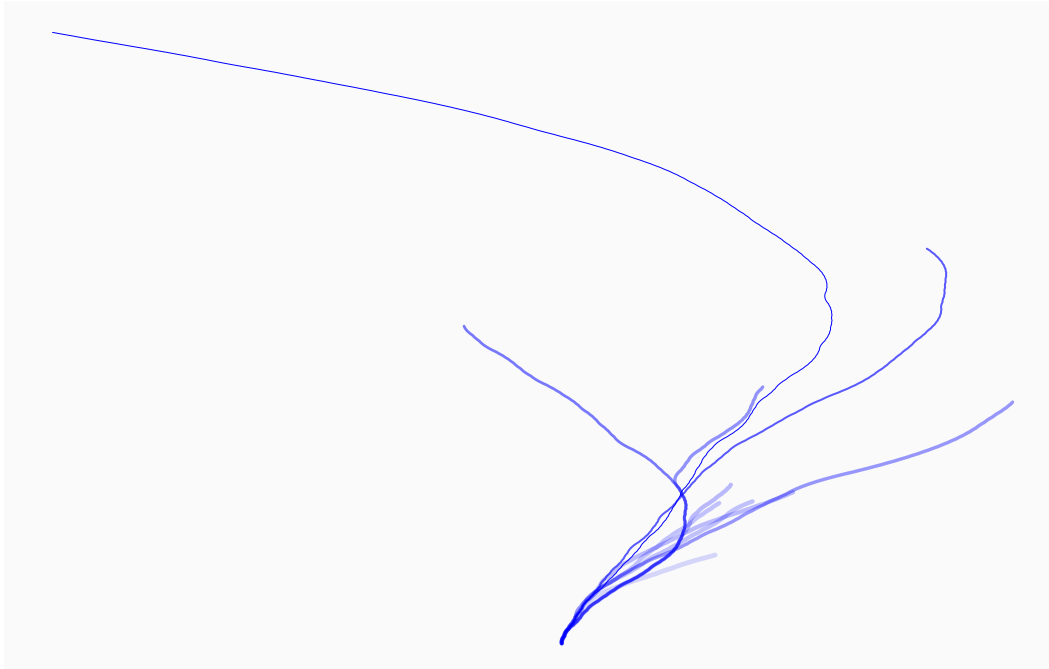
$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \equiv 0 \pmod{2} \\ \frac{3n+1}{2} & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

The Collatz conjecture, or the $3n + 1$ problem, claims that no matter which positive integer you start with, you will eventually reach 1.

Question 3

a) Canvas_collatz()

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



This ggplot shows how many steps (or the general path) it takes for a set of starting integers ($n = 10$) to reach 1. It's made to be plotted in blue.

b) Different Output?

- The function randomizes the 10 integers you start with. So, you could end up with a completely different plot each time you run it.
- This means it might look different and inconsistent each time you render it.
- With `set.seed()`, you use the same random set of integers associated with the seed number you provide (freezing or locking it in place), so you get the same result when you render it.

```
set.seed(9)

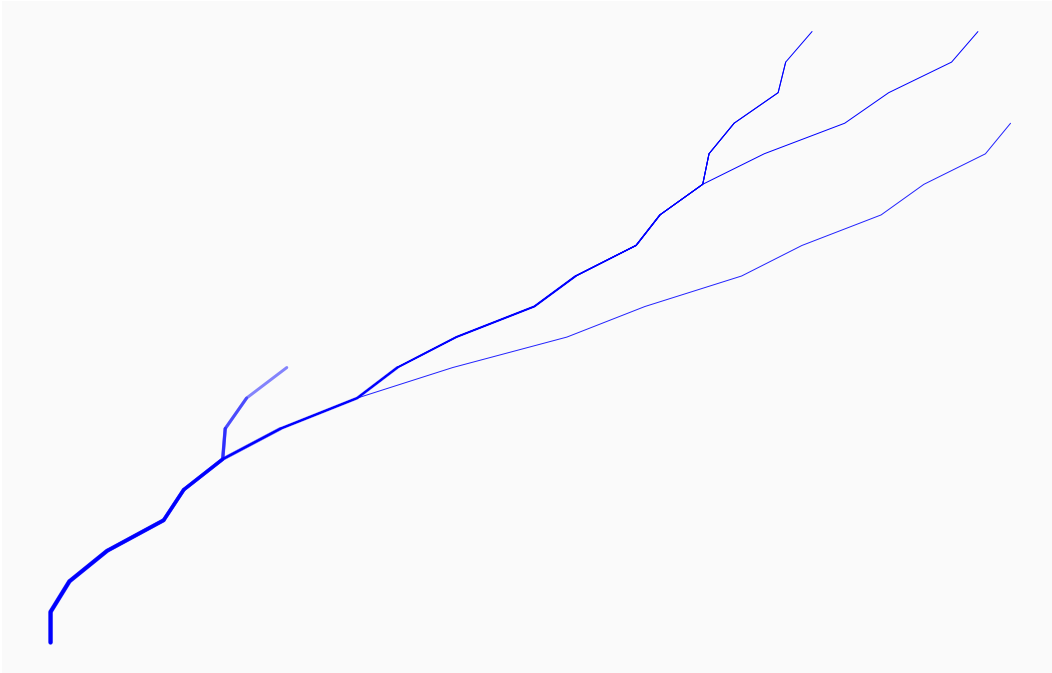
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



Question 4

```
set.seed(1)

canvas_collatz(
  colors = c("blue"),
  n = 1:20
)
```



```
canvas_collatz(
  colors = c("blue"),
  n = 1:20
)
```



Let's say you use a set of integers associated with seed 1. Even if you run the code without setting a seed, it produces the same plot. This is because $n = 1 : 20$ overrides the seed. You are essentially telling R to use the exact integers 1 through 20 as the starting values, instead of random ones.